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# Analysis of VPN Encryption Techniques, Vulnerabilities and Behaviours in Cloud and On-Premise Environments

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# Abstract

The rise in the number of clients using a Virtual Private Network (VPN) is significant, especially after the COVID-19 pandemic. This is reflected in the corporate world where numerous organisations now offer the possibility of working remotely. As a result, it is crucial for data to remain confidential while in transit. The solution to be able to achieve this is the use of a VPN. However, this technology still has vulnerabilities such as DoS and MiTM attacks which are addressed through this research. The latter is carried out mainly by analysing data in academic papers related to the field and also comparing exiting research and solutions.

This work demonstrates how different encryption algorithms and authentication mechanisms perform in different VPN tunneling protocol such as Generic Routing Encapsulation (GRE), Internet Protocol Security (IPSec) and Layer 2 Tunneling Protocol (L2TP). Analysis is carried out to assess the attacks which target VPNs and how can they be mitigated. It is equally seen how there exists a trade-off between security and performance. The mechanisms present in traditional networking VPNs and cloud-based VPNs were also studied. It is seen how using the cloud to deploy a VPN can provide benefits such as being more scalable, more resilient and lower costs. It is, nevertheless, noted that there are also security risks present while using this technology which do not exist when using traditional networking VPNs. This work finally suggests how organisations can use VPNs at lower costs as fewer physical resources are required if the cloud is used and at the same time illustrates how security is guaranteed with minimum loss in performance through the various security mechanisms in different environments.

# List of Acronyms

|                |                                                                    |
|----------------|--------------------------------------------------------------------|
| <b>VPN</b>     | Virtual Private Network                                            |
| <b>PPTP</b>    | Peer-to-Peer Tunneling Protocol, Point-to-Point Tunneling Protocol |
| <b>MiTM</b>    | Man-in-the-Middle-Attack                                           |
| <b>DNS</b>     | Domain Name System                                                 |
| <b>GRE</b>     | Generic Routing Encapsulation                                      |
| <b>IPSec</b>   | Internet Protocol Security                                         |
| <b>L2TP</b>    | Layer 2 Tunneling Protocol                                         |
| <b>PSK</b>     | Pre-shared key                                                     |
| <b>AH</b>      | Authentication Header                                              |
| <b>IKE</b>     | Internet Key Exchange                                              |
| <b>ESP</b>     | Encapsulation Security Payload                                     |
| <b>SA</b>      | Security Association                                               |
| <b>LAC</b>     | L2TP Access Concentrator                                           |
| <b>LNS</b>     | L2TP Network Servers                                               |
| <b>DoS</b>     | Denial of Service                                                  |
| <b>EAP</b>     | Extensible Authentication Protocol                                 |
| <b>MS-CHAP</b> | Microsoft Challenge Handshake Authentication Protocol              |
| <b>CHAP</b>    | Challenge Handshake Authentication Protocol                        |
| <b>DDos</b>    | Distributed Denial of Service                                      |
| <b>MFA</b>     | Multi-Factor Authentication                                        |
| <b>SDN</b>     | Software Defined Networking                                        |

**NFV** Network Function Virtualization  
**MPLS** Multi-Protocol Label Switching  
**AES** Advanced Encryption Standard  
**MD5** Message Digest Method 5  
**QoS** Quality of Service  
**IPS** Intrusion Prevention System  
**OSPF** Open Shortest Path First  
**LSA** Link State Advertisement  
**IDS** Intrusion Detection System  
**DES** Data Encryption Standard  
**ACL** Access Control List  
**RTT** Round Trip Time  
**bps** Bits per second  
**ARP** Address Resolution Protocol  
**ALPACA** Application Layer Protocol Content Confusion Attack  
**TLS** Transport Layer Security  
**SSL** Secure Socket Layer  
**DH** Diffie-Hellman  
**SSTP** Secure Socket Tunneling Protocol  
**SGX** Software Guard Extensions  
**HMAC** Hash Message Authentication Code  
**CBC** Cipher Block Chaining  
**ISAKMP** Internet Security Association and Key Management Protocol  
**UDP** User Datagram Protocol  
**RADIUS** Remote Authentication Dial-In User Service  
**NAT** Network Address Translation  
**VPNaaS** VPN as a Service  
**RSA** Rivest-Shamir-Adleman

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# Chapter I

## Introduction

### I.1 Keywords

VPN, tunneling protocols, IPSec, encryption, cryptography

### I.2 Introduction and context description

The Virtual Private Network (VPN) technology saw its days around the year 1996 when the peer-to-peer tunneling protocol (PPTP) was developed by a Microsoft employee. When VPN was recently introduced, it was used mainly by companies to give faster and flexible networking system to their employees as they had to ensure secure communication [44]. It is only with time that home users began to go on the Internet through a VPN as it provides better security online. Before the introduction of VPNs, organizations were using leased lines to establish a secure communication channel but the main disadvantage was its high cost. VPNs allow remote connections over a third party network which is the Internet thus reducing cost which is one of the reasons why this technology is so popular [36] .

VPNs are nowadays largely used by organizations to allow their employees to work remotely. This permits them to log safely into a company's network, enlarge their networking infrastructure and lower their costs [43]. This technology is equally used by individuals at home who wish to have access to services which are not available, or which are censored in their country [18] or even to have better security while connecting to the Internet [10]. The VPN market expanded to 30 billion dollars in 2020 when it was around 16.5 billion dollars in 2016 and according to estimates, this figure is anticipated to increase at a yearly rate of more than 15 percent between 2021 and 2027 [13]. A VPN offers several security mechanisms including encryption, tunneling, authentication and firewall to filter traffic. It is complex to have a precise definition of the subject matter but a technical one would be to create or expand a private network by passing through a public network which is typically the Internet. This allows for information to be shared by users in a particular group and it would look as if all activities on their devices are being executed on the non-public network itself although this is not actually the case. This illusion is the backbone behind the increase in demand for

security for services that are accessed through a VPN [17].

The main reason why VPNs are used is security and privacy. A VPN encrypts data while sending it on the Internet which makes it complex for the monitoring of online activities especially when dealing with public networks such as free Wi-Fi. Using this technology allows users to stay anonymous on the Internet and safety can be increased online. However, this is a double-edged sword as it can also be used for wrong purposes. Numerous users access resources which are restricted and sometimes illegal in their home country through a VPN as this changes the IP address of the device; making it seem like they are in another location. Yet, from a company's point of view, a VPN is used to create a secure tunnel on the Internet between the organisation and its employees who are working remotely. It is also better in terms of cost when compared to operating a private WAN to allow communication with the company's resources [17].

VPNs, however, are still exposed to several vulnerabilities such as Man-in-the-Middle (MITM) attacks, DNS Hijacking, trojans and worms [7], and even encryption cracking. It is true that this technology provides an additional layer of security for packets being sent over the Internet but as they are transmitted through a public network, they can be intercepted at any time. This project, therefore, tries to implement strong cryptography inside the VPN tunnels by analysing various tunneling protocols and the encryption algorithms being used such that communication is safer online which shall be done without any compromise in the speed of the network.

### 1.3 Problem Statement

More and more people are connected to the Internet through various devices, so, it is crucial that networks are secured. There has been an exponential increase in the number of people working remotely during the COVID-19 pandemic and even after it, many people are still working in this style. This is because organizations had to keep working to ensure that the needs of their customers were being met adequately even during the pandemic [3]. An interesting fact to note is that internet traffic escalated by up to 90 percent during the COVID-19 pandemic thus pushing companies to adopt the work-from-home style, which is one of the main factors leading to a rise in the number of people working remotely [7]. Sensitive information of companies offering employees the opportunity to work out of the office is highly exposed to risks as it is being sent openly through the Internet where there is little to no assurance of security.

### 1.4 Aims

This project explores how the various components of VPNs are used to ensure confidentiality, protect against data spoofing and protect sensitive data being transmitted through the Internet. The use of advanced encryption techniques to tackle data snooping coupled with the implementation of integrity checks [5] is also analysed.

## 1.5 Objectives

- Research to analyse existing tunneling protocols used in VPNs.
- Write project proposal and literature review.
- Research about the encryption mechanisms used in various tunneling protocols.
- Evaluation how the research questions have been answered.
- Analysis of encryption and authentication mechanisms in VPN in different platforms, that is, cloud-based VPNs and traditional networking VPNs.
- Deliver the project

## 1.6 Research Questions

- What are the different types of encryption used in various tunneling protocols such as GRE, IPSec, L2TP employed by VPNs?
- Is certificate-based authentication or a Pre-Shared Key better for security in IPSec?
- What are the attacks which target VPNs and how can they be mitigated?
- How passwords can be acquired and how traffic can be decrypted in L2TP/IPSec?
- How security in VPNs can be achieved with minimum loss in performance?
- How traditional networking VPNs and cloud-based VPNs work?

## 1.7 Ethics considerations

After filling the research ethics screening form, ethical approval is not being considered for this project as due to its nature, it does not require user evaluation and other factors which may need ethical consideration. The form can be found in Appendix A.

## 1.8 Deliverables

- Project proposal.
- Analysis on existing VPN tunneling protocols and encryption mechanisms used within tunnels
- Literature review
- Research to answer the research questions and attempt to contribute to literature of the topic.
- Final report
- Presentation

## **1.9 Resources**

- Academic papers
- Academic Journals
- Laptop

## **1.10 Summary of Chapters**

This report performs an extensive literature review regarding the tunneling protocols used in VPN technology to provide a general overview on the topic. Analysis is chiefly carried out on algorithms such as GRE, IPSec and L2TP. The existing research which have been carried out in the same direction are also presented. The existing solutions are also laid out and the works are compared against each other in order to address gaps in research. The next chapter would then be focusing on the research methodology and the results acquired will be presented. The research methodology will provide further insights of how the research has been carried out in a precise way and what was the scope of the work. It will also be seen how the results collected will later be assessed. This chapter also presents the key findings in a concise way and will be more specific towards answering the research questions in terms of the security of VPN in different platforms. This is then followed by the discussion of the findings which will consist mainly of the analysis of the research questions. This would involve analysing the encryption and authentication mechanisms as well as vulnerabilities present in VPNs on different platforms. This will then give a comprehensive analysis of the results discussed before moving to the final chapter which is the conclusion and future works. This final chapter has as task to do an analysis of the project which will outline the main findings of this work and the contribution in the field of VPN security on various platforms. The limitations of the project are also pointed out and directions in which future works can be carried out are stated so that it can be of benefit to the field of VPN technologies.

## Chapter 2

# Literature Review

### 2.1 Introduction

VPNs are used to maintain anonymity online and to stay safe and companies use this technology to allow sensitive data to be protected. This is mainly in situations where employees work remotely as data would be sent over a public network, that is, the Internet. There are various mechanisms which allow for this to be possible and these shall be explored. There are VPN tunneling protocols such as GRE, IPSec and L2TP which shall be explored in depth. A more comprehensive view of the issues in VPN will be analysed before looking at how the problems and the recommendations will impact communication in VPN channels. There are already several research and solutions which are closely related to this project and a comparative analysis of other solutions would be performed as well. However, there are very few works which address the various encryption mechanisms used by VPN tunneling protocols. Research involving authentication techniques such as using pre-shared keys or certificate-based authentication have not yet been exploited completely and there are also less focus on the attacks to which VPNs are vulnerable in existing literature. Thus, a more extensive review of literature will be done so that the gaps in these areas can be identified and further research can be carried out.

### 2.2 VPN Tunneling Protocols

Organisations need to protect their network infrastructure from cybercrimes and one way to achieve this is through the use of a VPN. The latter creates a communication tunnel between the private network and the Internet so that data can be transmitted securely between end devices. In this sense a VPN may be said to extend a private network to ensure confidentiality of data and this is done by using a set of various protocols available such as encryption techniques used to prevent unauthorised entities from having access to sensitive information [14]. These protocols would be introduced in this section to have a better understanding of how this technology operates. As seen above, protocols such as IPSec, GRE and L2TP shall be analysed and these include both site-to-site and remote access VPNs.

### 2.2.1 GRE

In the GRE protocol, the tunnel technology between protocol layers is used where point-to-point connection is supported in the tunnel. At both ends of the tunnel, the packets can be encapsulated and decapsulated [22]. In other words, GRE allows an encapsulated datagram to travel on another network layer protocol such as the Internet Protocol [40]. Before unpacking a packet to send it to the desired location, there is a process which should be followed. The GRE header and tail of the initial data packet are encapsulated and then a new IP header is added to transmit the encapsulated packet from the GRE source interface to the other end of the tunnel [22]

To transmit data through a public network in a fast, reliable way and to allow for easy communication, the Generic Routing Encapsulation (GRE) was used. A secure and reliable path for communication is created by using a point to point private connection. However, this protocol is not as secure as the IPSec protocol due to the fact that advanced security mechanisms such as encryption, authentication and sequencing are not present [15]. In the sender part of the tunnel, the last step of GREs in the routers will acknowledge the data packet before encapsulating it with the GRE header and the destination address of the tunnel. When it comes to the recipient side of the tunnel, the packet is sent to the destination after the receiver end routers have decapsulated the packet received [2].

### 2.2.2 IPSec

Internet Protocol Security (IPSec) refers to a suite of protocols which includes three main ones which are the Authentication Header (AH), Internet Key Exchange (IKE) and Encapsulation Security Payload (ESP) [40]. It can clearly be observed that it is more secure than GRE when it comes to tunneling protocols in VPN and it can be used in site-to-site access VPN as well as remote access VPN [15]. This protocol has two modes namely tunnel mode and transport mode. The main components of this protocol will be analysed.

AH is used to allow for authentication and data integrity. It also provides features such as replay protection and access protection. However, AH does not guarantee data confidentiality on its own. Hence, it is used together with ESP to ensure privacy. ESP works by performing a cryptographic function on a packet's payload and symmetric encryption is used in this case so that both IPSec ends can use only one key to encrypt or decrypt the packet [31]. The IPSec tunneling creates a secure path between different ends which need to communicate. This process is done through the Internet Security Association (SA) [15]. To negotiate, define and manage this process are the duties of the IKE. Consequently, to benefit from a secure tunnel through IPSec connections for communication, the SA has a series of approved keys, security parameters and protocols [31].

### 2.2.3 L2TP

Layer 2 Tunneling Protocol (L2TP) is a remote access VPN tunneling method used. However, it does not contain strong security mechanisms such as authentication, encryption, confidentiality and integrity for each packet as compared to IPSec [39]. As the latter is a layer 3 protocol, tunnels for layer 2 packets cannot be created only by IPSec. This is why L2TP is

used together with IPSec so as to guarantee the security of each individual packet [15].

The L2TP/IPSec tunnels have two ends of communication which include the L2TP Access Concentrator (LAC) and L2TP Network Servers (LNS) where the former acts as a client and the latter as a server. Thus, to connect remote users, a client-server approach is used. The certificate-based authentication is already configured both on the client and the server. To provide a robust encryption and authentication, the IKE is determined using the certificate-based authentication while the security settings for L2TP are being arranged [39].

## 2.3 Problem Description

As seen above, the use of VPNs has increased as many firms offer their employees the option to work remotely. This, however, means that the employees should have access to the company's sensitive data without being on premises which implies that information is being transmitted through the Internet where there are numerous vulnerabilities present. No third parties should be able to access this data and even if intercepted, it should not be readable to unauthorised entities. One way to ensure this is through the use of encryption in secure VPN tunnels as discussed above.

There still exists entities which refuse to deal online due to the various security threats associated with this way of operating even through a VPN whereby all information are passed within a secure tunnel but their apprehensions are fully justified as there may still be vulnerabilities in the system thus exposing crucial data out in the open. There are 479 vulnerabilities which have been discovered till now and among these there are the Top 28 exposed in the year 2020 alone [7]. VPNs are also exposed to attacks such as Denial of Service (DoS) attacks, Distributed Denial of service (DDoS) attacks, Man-in-the-Middle (MiTM) attacks, DNS Hijacking, trojans and worms [7] against which a secure VPN shall be protected from. It is, however, also important to have an appropriate speed when using VPNs and low latency due to the critical tasks which are performed through this technology such as transmission of confidential data of companies and personal bank details of individuals. This is why strong security should also be present in the VPN tunnel alongside performance.

There are various number of research which have been carried out but due to the speed at which technology is progressing, many of them are outdated and have not been adapted to the needs of technology used nowadays. For instance, VPN solutions were tested for only a few encryption algorithms and more emphasis was laid on performance [20]. However, these solutions would be risky to use now due to the rapid changes in technology. Thus, there is a need for more emphasis to be laid on different encryption mechanisms used in various VPN tunneling protocols to guarantee security. For instance, in IPSec technology, encryption of the data payload is ensured by the ESP which enables data confidentiality and integrity[9]. The L2TP uses Frame Relay, X.25 networks and Asynchronous Transfer mode to transmit data between systems. The point-to-point protocol uses authentication mechanisms such as EAP, MSCHAP and CHAP which are equally present in L2TP. However, data confidentiality is not offered by L2TP and hence, it is combined with IPSec to form the L2TP/IPSec to provide so [8]. It will also be seen how VPN providers can improve their authentication

mechanisms by using certificate-based authentication rather than pre-shared keys to ensure integrity of data. Research will be carried out equally on the attacks which target VPNs such as Denial of Service (DoS) or Distributed DoS (DDoS), Man-in-the-Middle attacks (MiTM), replay attacks amongst others.

It is important to note that VPNs are not fully secure and that there are still attacks which may compromise VPN tunnels. For instance, there are attacks which can be carried out to decrypt traffic in the L2TP/IPSec protocol and also recover the passwords in use for authentication purposes. It will be seen how this may occur and how it may be prevented as well. Many entities are reluctant to move to VPNs which provide better security partly under the excuse that doing so will result in a decline in the performance of the service. Thus, analysis will be performed on how security can be ensured in VPNs with minimum compromise of the performance. A relatively new term in the networking field is 'cloud computing'. VPNs can also be established using either traditional networking or cloud-based VPNs. Research is carried out to better understand what are the underlying mechanisms used by each type so as to ensure security.

## 2.4 Sources and Impacts

In this digital age, it is crucial for data to be safe online and this can be achieved by through a VPN. The latter can be used mainly to protect from cyber crimes on networks with weak security and even maintain anonymity and remain safe online [27]. It can be seen that in the paper [23], VPN is used for reasons such as remote office which involves transmitting sensitive data over a public network but through a secure tunnel. There are works such as [30] which outlined how DoS attacks can be performed on VPNs and thus, putting crucial information at risk.

However, there are also solutions which have been proposed in academic papers to tackle the issue of security in VPNs. For instance, authors in [4] explained how using certificate-based authentication is better than using pre-shared keys. The paper [34] analysed how encryption mechanisms may affect computer networks. Combining several ideas from a range of papers, it was seen that a VPN with security features such as advanced encryption algorithms, firewall and MFA shall be developed. By achieving this, the transmission of data in VPN tunnels would be made more secure and thus, more organisations would be more open to allow remote working for their employees as communication would be more secure.

## 2.5 Existing Research

- **Security Assessment and Evaluation of VPNs: A Comprehensive Survey [1]**

This survey pointed out that users do not make the most of VPN technology due to problems associated to performance, security and configurations. This work analysed the types of VPNs and compared different features such as encryption, key length, privacy and security for various VPN protocols. Analysis was also done on security and



performance related problems in VPNs. Discussion was also made on how a VPN can be selected depending on the needs of an organisation and how they can be deployed securely. The security risks and attacks associated to them were also outlined. The performance-related facet was also explored. This includes the performance and scalability, packet retransmission handling and OpenVPN support for security reasons. It was also seen how auditing practices such as security policy, firewall rule base and log settings should be implemented. Some directions for future works were also put forward in this paper which include IPv6 network testing, application of Software Defined Networking (SDN) in VPNs amongst others. This research analysed various issues related to the security and performance of VPNs but it could have also explored the cloud-based approach for the implementation of VPNs which could have provided different insights.

- **A study of cloud-based VPN establishment using Network Function Virtualization technique [28]**

This work studies how users can benefit from the security components of VPNs through network function virtualization (NFV). This term refers to shifting networking features such as load balancing, firewall, antivirus from dedicated hardware to a virtual server, that is, the cloud. This research made use of SDN and NFV as the main mechanisms to propose a cloud model for VPN. The design included three main elements namely the forwarding plane, signaling and control plane and data plane where each one has a specific function. It was also seen how Multi-Protocol Label Switching (MPLS) is used in VPNs using traditional networking and the high costs associated to it being one of the reasons encouraging VPN providers to adopt a cloud-based approach for the deployment of VPNs to users. The model illustrated how a cloud-based VPN creates communication links and overheads and complexity is less when compared to traditional VPN networking. However, there are also multiple security risks associated when adopting a cloud-based approach for VPN as services would be managed by a third-party organisation. The paper could have explored this aspect as well and try to propose solutions to tackle them.

- **A research on the impact of encryption algorithms on the quality of VPN tunnels' transmission[34]**

There have also been works where VPN tunnels which supported cryptography methods such as AES and hash mechanisms such as MD5 and SHA were designed to analyse the effects of encryption and hashing algorithms on the Quality of Service (QoS) among connected devices. Nevertheless, there are factors such as IPSec VPN, firewall or Intrusion Prevention System (IPS) which may lead to a decrease in performance [34]. Therefore, it has been seen that there has been a trade-off in the ease of use and information security in the research in this work. Thus, there should be works which attempt to develop a secure VPN transmission channel without putting user experience at risk.

## 2.6 Existing Solutions

- **Dos vulnerability verification of IPSec VPN [45]**

This work studies how the IKE protocol is vulnerable in the IPSec VPN key exchange protocol. This paper also demonstrated how there is DoS attacks risks in IPSec VPN. The primary protocol to route traffic between a client and a VPN server is the Open Shortest Path First (OSPF). The work illustrated how the weaknesses in the Link State Advertisement (LSA) can be exploited by a DoS attack [30]. The research made an extensive analysis into how DoS attacks can be launched in IPSec VPN application. This paper illustrated why it is crucial to have a secure key negotiation process in IPSec VPNs so as the confidentiality and integrity of data transmitted over a public network is not compromised [4]. However, it could be improved by exploring the countermeasures to prevent such kind of attacks which may include systems such as Intrusion Prevention Systems (IPS) or Intrusion Detection Systems (IDS). There are candidates such as Suricata and Snort which can be used for this as they also have low overheads [30]. While this work provides an extensive analysis of the risks of IPSec VPNs, further research into particular countermeasures could have contributed to how these threats could be addressed.

- **VPN remote access OSPF-based VPN security vulnerabilities and counter measurement [30]**

This research explored how the number of clients using VPNs increased drastically through the COVID-19 pandemic and how the flaws in this technology easily became target points to be exploited by attackers. The weakness of key negotiation process in the main mode and the aggressive mode of Internet Key Exchange (IKE) protocol in IPSec VPN is treated in this paper. A DoS attack based on the OSPF protocol adjacent route spoofing was tested in this work. The paper [45] also lays emphasis on the vulnerability in the process of key negotiation in the main and aggressive mode of IKEv1 protocol in IPSec VPN. The work then suggests a DoS attack based on the adjacent route spoofing of the OSPF protocol. Authors in [33] analyse a range of stateless flooding attacks and implemented multiple attacks to evaluate the DoS resilience of four mainly used VPNs. The research illustrated why there is a need for adversarial testing and why it is necessary to make VPNs more resilient to DoS attacks. A solution to protect the VPN from DoS attacks was proposed in [30] which involves using the Suricata as an IDS. The paper does not only analyse the main vulnerabilities in the IKE protocol in IPSec VPN but also proposed a solution to tackle potential DoS attacks which may arise. This implies that this research can be helpful to enhance the security of VPN.

- **Cross protocol attack on IPSec-based VPN [4]**

The paper carried out an analysis of the IPSec based open-source VPNs and how security might be compromised if vulnerabilities present are exploited by attackers. This work laid great emphasis on how the use of certificate-based authentication is more

secure for IPSec instead of using a Pre-Shared key for authentication. The paper attempted to show a cross protocol attack on IpSec-based VPN but was unable to do so. However, it concluded that IPSec is robust enough when it comes to the strong security features and the strong authentication mechanisms. The solution proposed had weaknesses in key management and authentication. There are other research on cross protocol attacks such as [6] where the target protocols are TLS/SSL. This paper demonstrates the vulnerabilities which exist in SSLv2. Nevertheless, the work could have been improved by employing features such as multi-factor authentication (MFA) and advanced encryption mechanisms such as AES which could have demonstrated how IPSec VPNs may not be fully secure. The research also only tested that a MiTM attack cannot be performed before concluding but there are other kinds of attacks such as DoS and Replay Attacks which may still occur on a VPN. Thus, IPSec VPNs are still vulnerable when it comes to cyber attacks despite there are already some highly secure technologies employed. A more in-depth evaluation of IPSec VPN security could have therefore, been performed if a wider range of cyber threats were addressed in this paper.

## 2.7 Comparative Analysis

In this section, various related research to the topic will be analysed and compared. This will be done in terms of the environment used in the work, why the research exist, how they were implemented and observations on what could have been improved. The table below compares the technical aspects of some research carried out in the field in terms of how they work, their environments and the architecture used in different works.

| Authors                    | System Test Environment       | Description                                                                                                                                                                                                                                                                                         | Architecture                                                                                                                                                                                                                         |
|----------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zhou and Zhang, 2020 [45]  | GNS3                          | There is a VPN server and client which both use IKEv1 for negotiation. The pre-shared key authentication mode is used when adopting the aggressive mode. For encryption mode, aes-cbc is used and hmac-sha128 as algorithm used for verification. Finally, dh-1024 is used for key-exchange.        | There are two PCs in the server and client intranet where the IPSec VPN which have been established as gateway routers for communication. There is also an attacker which is an Ubuntu 16 system virtual machine.                    |
| Sawalmeh et al., 2021 [30] | GNS3, Ubuntu virtual machines | IPSec and IPSec VPN IKEv1 settings have been used for negotiation and network connection.                                                                                                                                                                                                           | There are 10 routers and the connection among them is determined through the OSPFv2 protocol. The Ubuntu machine illustrates both the client and the server. The remote access type VPN is used to define the VPN configurations.    |
| Sankaran, 2023 [4]         | VirtualBox, Wireshark         | The IPSec implementations using strongSwan, OpenSwan and LibreSwan were installed and configured in the server VM so as to create an IPSec VPN tunnel between the desktop and server VMs. To intercept the network traffic between the hosts and to test the IPSec VPN tunnels, Wireshark was used. | A server and a desktop was connected through an IPSec tunnel in a virtual environment. This was done by creating a virtual machine for the server which is Ubuntu server 20.04 and one for the client which is Ubuntu desktop 20.04. |
| Xu and Ni, 2020 [42]       | OpenSwan                      | OpenSwan is used in this work to develop a simple IPSec VPN network under Linux operation.                                                                                                                                                                                                          | The VPN tunnel is established using two virtual machines for connection.                                                                                                                                                             |

Table 2.1: Comparative analysis 1

The table below illustrates a more in-depth comparison of existing solutions and observations have also been made.

| Authors                        | System Test Environment       | Use/Reason of Existence                                                                                                                                                          | Implementation                                                                                                                                                                                                                                                                                                                                | Observations                                                                                                                                                                                                                                       |
|--------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zhou and Zhang, 2020 [45]      | GNS3                          | This work explored how the use of IKEv1 protocol is not fully secure on IPsec VPN.                                                                                               | This paper experimented a DoS attack based on the OSPF protocol adjacent route spoofing. GNS3 software was used to create a simulation network topology so as to verify how IKEv1 is vulnerable to DoS attacks.                                                                                                                               | The work could have explored the countermeasures to treat the threats analysed. This could have helped in dealing with attacks which target VPNs and thus, improve its security.                                                                   |
| Sawalmeh et al., 2021 [30]     | GNS3, Ubuntu virtual machines | This research employs techniques to protect a VPN against DoS attacks by using the Suricata as an IDS.                                                                           | This work used mainly the environment simulated by GNS3 for its experiments and to perform basic configuration operations. It also proposed two solutions to prevent a DoS attack on a VPN server. They are namely checking the source port to allow or deny a client connection and integrating strong IDS/IPS tools into the communication. | Both solutions proposed in the research will make use of an extensive amount of resources and so the connection to the VPN server will be slow for the clients. This may discourage users to adapt a secure VPN due to the decline in performance. |
| Sankaran, 2023 [4]             | VirtualBox, Wireshark         | This paper shows why it is better to use certificate-based authentication instead of a Pre-Shared Key for authentication so as to strengthen the security of IPsec.              | A virtual network was created with a server and a desktop connected through an IPsec VPN. To do so, VirtualBox was used to install the virtual machines (VMs) and Ubuntu was used in this case. For the network configuration the two VMs were assigned static IP addresses so that there is a reliable connection.                           | This work had vulnerabilities in authentication and key management. Thus, it demonstrated why it is better to use certificate-based authentication rather than pre-shared key.                                                                     |
| Strzeciwillk et al., 2018 [34] | -                             | This work illustrated the impacts of the use of encryption algorithms on a computer network.                                                                                     | This work implemented and tested a network model which uses the AES encryption algorithm and the hash functions MD5 and SHA in VPN tunnels. The study was mainly about how varying data lengths affected the quality of transmission in a secured network.                                                                                    | The user experience when applying the encryption techniques in this work was compromised. There should be research about how to develop a secure VPN channel which does not affect user experience to a large extent.                              |
| Luo and Ji, 2020 [24]          | -                             | This work analyses how the encryption mechanisms used in L2TP/IPsec might be vulnerable to password acquisition and traffic decryption.                                          | This paper demonstrates how pre-shared keys and login passwords can be obtained in L2TP/IPsec tunnels using high-performance computing. It also shows how data can be decrypted through MiTM attacks.                                                                                                                                         | The research involved analysing vulnerabilities in encryption mechanisms in L2TP/IPsec and how data can be decrypted even in data tunnels. The cryptographic techniques used can be improved so as to prevent this.                                |
| Xu and Ni, 2020 [42]           | OpenSwan                      | This paper carries out research about how to ensure Internet security when using VPN.                                                                                            | This research used OpenSwan to create a simple virtual VPN tunnel between two virtual machines. An IPsec VPN tunnel was created through this software under Linux operation.                                                                                                                                                                  | There can be improvements in this work in terms of better management of time and resources, the throughput of the system and better handling of data packets.                                                                                      |
| Liu, 2023 [23]                 | -                             | This paper studied how VPNs work, the importance of this technology for uses such as privacy protection, remote office and geographical restrictions.                            | This research used real case studies to explore how popular VPN providers employ their security mechanisms. It also points out the VPN data breaches which occurred in earlier so as to guide users how to pick a good VPN provider and also be aware of the risks.                                                                           | There can be further research in the fields of how user habits will impact VPN usage and how VPNs will perform in high-speed network environments.                                                                                                 |
| Abbas et al., 2023 [1]         | -                             | This work surveyed the features of VPN technologies based on their security, performance, auditing and management function before proposing a strategy to assess a VPN solution. | The paper does a comparative analysis of a series of VPNs such as NordVPN, TunnelBear, OpenVPN and Hotspot Shield in terms of performance tests, auditing and security.                                                                                                                                                                       | The research pointed out many factors which could be improved in VPNs to make it more secure but did not implement them.                                                                                                                           |

Table 2.2: Comparative Analysis 2

## 2.8 Summary of Findings

After various existing solutions and research have been analysed and compared, the points discovered shall be presented in a more concise way.

- There are various vulnerabilities in IPSec and even L2TP/IPSec protocols.
- Attacks were simulated to demonstrate weaknesses in VPNs and how security can be ensured.
- Security mechanisms such as IPS/IDS to secure VPNs against threats were analysed.
- Surveys pointed out threats in VPNs and how users can select a trustworthy VPN provider.
- There were, however, less focus on the security mechanisms such as encryption and hashing algorithms used to transmit data in VPN tunnels.
- Much emphasis was not laid on how security can be applied while ensuring that performance is least affected.
- There can be further research in terms of features such as implementing a firewall and multi-factor authentication (MFA) in VPN tunnels.

## Chapter 3

# Research Methodology and Results

### 3.1 Research Methodology

#### 3.1.1 Research Design

The research for this project follows a mixed-methods approach which includes both quantitative and qualitative research. For instance, it analyses the performance of a VPN under different circumstances and also the architecture of the various types of VPNs. There is an attempt to find how security can be ensured in VPNs but at the same time, there should be minimum loss in performance. This is done by analysing the different types of encryption mechanisms used in VPNs, deciding on secure authentication methods and also studying the attacks which can be performed on VPNs. To be able to achieve this, there has been the use of academic papers and the key search criteria were, 'VPN Security', 'IPSec VPN', 'Enhancing VPN security', 'Encryption mechanisms in VPN', 'Cloud-based VPNs', 'Traditional networking VPN', 'MPLS', 'VPN tunneling protocols', 'GRE', 'L2TP/IPSec'. The main repositories for research papers for this project are IEEEExplore, ACM Digital Library and Elsevier.

#### 3.1.2 Meta-analysis

A thorough meta-analysis is carried out based on existing VPN tunneling protocols explaining how they work and provide security in data transmission. It is observed that after the COVID-19 pandemic, many organisations have adapted the remote work approach thereby making use of VPN technologies. However, it should be pointed out that there are security threats to this technology as VPNs are still vulnerable to attacks. Thus, research is carried out in the direction to see how security may be improved in VPNs.

### 3.1.3 Scope of Research

This research deals with various security mechanisms present in VPN tunneling protocols. For example, the encryption techniques used in each VPN, how authentication is done, the attacks against VPN and it also covers the aspect of cloud-based VPNs. This is achieved by analysing the work performed in particular papers such as where research has been carried out on the security assessment and evaluation of VPNs; how encryption algorithms influence the performance in a VPN tunnel amongst others. The solutions presently available to improve the security of VPNs are also analysed and compared. Data is collected for this work and this is done by studying existing frameworks from academic papers and discussing how the security of VPNs can be enhanced within the scope of this project.

### 3.1.4 Evaluation

Security forms part of the major factors which contribute to business success, together with speed especially when it comes to sending data through an unsecured and shared network [14]. The main procedure followed for the results analysis phase is standardised experiments to assess the various components of a robust VPN. The results are mainly assessed on how the research questions have been analysed. For instance, it is seen how different encryption algorithms provide various levels of security; how an authentication method may be more secure than another and what techniques can be used to mitigate attacks against VPNs. Thus, the whole project can be evaluated on the grounds of how the research questions may have contributed to research in the field of VPN security.

## 3.2 Results Analysis

### 3.2.1 Key Findings and Results

In this section, data which have been diligently collected from different sources including case studies and standardised experiments are presented in a concise manner. Emphasis is laid on the various algorithms utilised by various tunneling protocols as well as different security measures. These results are being analysed especially to observe how security measures can be implemented while impacting the performance of VPN technologies to a minimum amount. The attacks to which VPNs are exposed to are also mentioned and an overview of how cloud-based VPNs and traditional VPNs work is pointed out.

The table below shows the key results which have been obtained for the various components of this research. This includes findings about GRE, IPSec, L2TP, certificate-based authentication compared to Pre-Shared key authentication, attacks on VPNs and cloud-based VPNs compared to traditional VPNs.



| GRE                                                                                                                        | IPSec                                                                                                                                                                                                                                                                                                                               | L2TP                                                                                                                                                                                                                      | Certificate-based authentication V/S Pre-Shared key authentication                                                                                                            | Attacks on VPNs                                                                           | Cloud-based VPNs V/S On-Premise VPN                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Encryption Mechanism: None. It is used with other protocols such as IPSec to provide encryption in data transmission [40]. | Encryption Mechanism: Authentication Header (AH), Encapsulating Security Payload (ESP), Internet Key Exchange (IKE) [31]. It uses DES-CBC, 3DES-SHA1, AES and Blowfish [9]. AH is used for authentication without encryption, ESP is used for encryption, integrity and authentication and IKE is the key management protocol used. | Does not provide authentication, encryption, confidentiality and integrity for individual packets [39]. It uses DES, 3DES, AES [9]. It is combined with IPSec to create the L2TP/IPSec protocol to provide security [39]. | Pre-Shared key authentication has been observed to be more vulnerable to cross-protocol attacks than certificate-based authentication when it comes to the security of IPSec. | Man-in-the-Middle (MiTM), Denial of Service (DoS), Replay Attacks, VPN tunnel compromise. | In traditional networking, MPLS is used to exchange label binding information and this is costly. These are also sometimes not scalable. However, cloud-based VPNs can prove to have lower cost and less build time and they are also more scalable, efficient, and resilient. |

Table 3.1: Key Overview on Research

The following sections represent the data collected from various papers which are in terms of frameworks and metrics which would be of use for this project as well.

### 3.2.2 Encryption Algorithms in Tunneling Protocols

As mentioned earlier, different VPN tunneling protocols have different encryption algorithms. Various works have been carried out to analyse how each of them is equally important depending on the requirements of an organisation. The paper [15] analyses the most used VPN application scenarios such as communication with branch office and also remote users. In this work, the simulation software GNS3 has been used to perform experiments for site-to-site and remote access VPNs. Both GRE and IPSec tunnels are configured but the focus here will be on IPSec. This configuration has two main phases where the first one deals with the encryption used (3DES), the authentication technique (pre-shared key) and the hashing algorithm (MD5) used to create the tunnel. The second phase involves applying security mechanisms such as Access control Lists (ACL). A point-to-point tunneling protocol was also set up using L2TP/IPSec to study encryption algorithms in remote access VPNs. The performance of the experiments performed is assessed using metrics such as throughput, Round Trip Time (RTT), jitter and security. Throughput can be said to be the speed at which messages are successfully being transmitted from a source to a destination in a specific amount of time and is normally measured in bits per second (bps). It is calculated by taking the TCP Window Size (b) and dividing it with latency (s). RTT refers to the time a packet takes to travel from one point to another and come back again [15]. In this case, a low RTT means that the performance is high. The term jitter is used to illustrate the variation in latency or response time which is measured in milliseconds. When security was evaluated, it was pointed out that data is only encapsulated within the GRE tunnel without encryption and authentication services. However, it was seen that IPSec provides a robust encryption and allow for authentication between two entities through the use of a pre-shared key where security associations are also created. L2TP/IPSec was found to be secure for PPPoE as it

combines the security services of IPSec together with those of L2TP [15].

To ensure security while using VPNs, there are authentication algorithms and encryption mechanisms which are used as seen above. It can be classified for each tunneling protocol as shown in the table below. The findings below represent a more technical approach as to how different tunneling protocols make use of different authentication and encryption algorithms.

| Tunneling protocol | Terminal Authentication                                         | User Authentication                                                                                                   | Encryption                                  |
|--------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| IPSec              | ESP/AH HMAC-MD5, IKE with digital certificate or pre-shared key | Digital certificates, shared secret passwords for mutual authentication.                                              | ESP (DES-CBC, 3DES-SHA1, AES and Blowfish). |
| L2TP               | IPSec ESP Computer Certificates.                                | PAP, CHAP, MS-CHAP, EAP-TLS digital certificates, Certificate Authority and pre-shared key for mutual authentication. | IPSec/ESP (DES, 3DES, AES).                 |

Table 3.2: Authentication and Encryption Mechanisms [9]

It has been seen that L2TP alone may not be the best choice. However, it can be used together with the IPSec protocol and thus from the protocol L2TP/IPSec. The technical components for this are as follows.

| Tunneling protocol | Encryption                  | Key Length | Speed                                                                    | Reliability and Scalability         |
|--------------------|-----------------------------|------------|--------------------------------------------------------------------------|-------------------------------------|
| L2TP/IPSec         | IPSec coupled with 3DES/AES | 256 bits   | Resources are needed for data encapsulation. Thus, relatively low speed. | Reliable for NAT supported devices. |

Table 3.3: L2TP/IPSec [1]

### 3.2.3 Authentication Mechanisms in VPN

VPN tunneling protocols use authentication algorithms to provide security in terms of integrity of data. IPSec uses mainly pre-shared keys through the use of IKE or ISAKMP which also serves in establishing a secure communication tunnel. To assess the effectiveness of this method, the paper [4] carried out experiments to simulate a cross-protocol attack on the VPN tunnel. A virtual environment was created and analysed using Wireshark. The attack was executed through a machine with Kali Linux OS which intercepted the packets. This was done chiefly through MiTM attacks with various approaches such as Address Resolution Protocol (ARP) poisoning and ARP spoofing. Nevertheless, it was discovered that these attempts were not successful because of the certificates used by the IPSec protocol and other security mechanisms present. This experiment was evaluated by analysing how security features in IPSec make it resistant to attacks. First, the authentication technique using RSA certificates proved to be effective in preventing the cross-protocol attack as it is complex for an intruder to carry out an impersonation attack hence improving the security of VPN. IPSec

works on the network layer which provides further security as it helps in preventing an attacker from intercepting and making changes in traffic being transmitted between VPN endpoints. The way IPSec VPN is implemented also differs from others such as the encryption and authentication techniques used which also help in ensuring security. For instance, experiments were carried out on the TLS/SSL protocol which runs on the transport layer where the Application Layer protocol Content Confusion Attack (ALPACA) cross-protocol attack was illustrated [4].

### 3.2.4 Attacks on VPN

The number of users for VPN technology have significantly gone up especially after the COVID-19 pandemic. However, this technology also contains vulnerabilities which attackers are constantly trying to exploit [30] and these include various attacks such as MiTM attacks, DoS attacks, replay attacks amongst others. The paper [33] studied DoS attacks focusing on stateless flooding attacks using a well defined framework. A test environment to simulate flooding attacks was created and the impact of the attacks were measured using the metrics throughput and time taken to set up the connection which refers to the amount of time for VPN hosts to set up a new connection. Several techniques to reduce such attacks were proposed such as source blocking, rate limiting and address-binding cookies. The paper [30] also analyses DoS attacks but it is based on the OSPF protocol adjacent route spoofing. The simulation software GNS3 was used to carry out experiments and evaluate the project. The solution proposed involved implementing an IDS/IPS software known as Suricata. This choice was made because malicious activity can be detected quickly with it and it can be controlled by the user depending on his specific needs. The work [10] pointed out the various vulnerabilities discovered in VPNs and also proposed mitigation strategies to deal with them. There was an experimental set up done in order to test two different types of attack namely network attack and local attack. The main vulnerability found in this study was the use of known pre-shared keys which could be used to carry out a MiTM attack on the VPN tunnel and intercept traffic.

### 3.2.5 Acquisition of Passwords and Traffic Decryption in L2TP/IPSec

It is of utmost importance to have data confidentiality present in a VPN tunnel for it to be secure. The paper [24] carried out experiments to explore the vulnerabilities present in L2TP/IPSec protocol and how passwords can be obtained and traffic can be decrypted through them. The framework used is mainly a MiTM attack carried out through high-performance computing proving that data can be intercepted and also understood by an attacker. The work put forward a cracking password to successfully obtain the login password by cracking the MD4 hash algorithm. However, the main weakness which allowed this is the mechanism used for key exchange for authentication purposes, that is, the pre-shared key. Traffic decryption was also performed through a MiTM attack and this was done after analysing the cryptographic mechanisms and encryption algorithms in the L2TP/IPSec protocol. This work deals with obtaining the encryption key of the Diffie-Hellman to crack data on the model proposed.

### 3.2.6 Trade-off between Security and Performance in VPN

It has been seen that numerous organisations are adopting the VPN technology to provide remote work services. Thus, it is crucial that sensitive data being transmitted in the VPN tunnels are secure as the vulnerabilities which might be exploited by attackers have also been pointed out. However, there may be many entities which are reluctant to make use of the most secure technology due to fears that it will lead to a decline performance of the VPN. The paper [1] carried out a survey and provided insights on how the performance of VPNs can be affected. For instance, it was pointed out that the performance will be affected differently on different operating systems. This was measured using the metrics such as delay, jitter and throughput in [21] for the protocol SSTP and IKEv2. Research carried out in [41] illustrates that it is the transport protocol which impacts the performance of VPN rather than the encryption algorithms for SSL VPN.

### 3.2.7 Traditional Networking VPNs and Cloud-Based VPNs

Network services can nowadays be provided either by on-premise equipment or by third-party providers which is known as cloud services. This is also the case for VPN services. The differences between the two types of VPNs are pointed out and there are also models proposed based on the cloud-approach in an attempt to improve this technology. The work [28] studied the establishment of a cloud-based VPN through the network function virtualization technique. This means moving network services from dedicated hardware to a third-party provider. This paper demonstrates the benefits of using cloud-based VPNs such as lower cost, efficiency and scalability. The work [25] lays more emphasis on the security threats which exist when using cloud-based VPNs such as key disclosure, impersonation and packet sniffing. The paper proposed a solution which is to use the EnclaveVPN which makes use of Software Guard Extensions (SGX) to ensure security. This model was evaluated by studying the performance of the enclave control plane and the network performance of the enclave data plane. The metrics throughput and latency were also used to evaluate the performance of EnclaveVPN on the cloud.

### 3.2.8 Comparison of Findings with Existing Literature

This section provides an overview of existing literature and how research was carried out in the field.

| Authors                                                                                       | Tunneling protocols analysed                          | Description                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sllame, Azeddien M, 2022 [31]                                                                 | MPLS VPN, IPSec                                       | Experiment is carried out for the performance of streaming applications using MPLS VPN and IPSec through OPNET tool. For testing purposes, HTTP and FTP traffics were also injected.                                                                                                                                                                                                          |
| Luo, Jie and Ji, Qingbing, 2020 [24]                                                          | L2TP/IPSec                                            | The vulnerability analysis of L2TP/IPSec protocol is performed and proposes an attack method for password acquisition and traffic decryption based on the protocol.                                                                                                                                                                                                                           |
| Santhanamahalingam, Sankili and Alagarsamy, Saravanan and Subramanian, Karthikeyan, 2022 [29] | Cloud-based VPN, Network as a Solution VPN (NaaS VPN) | This paper proposes a model for a cloud VPN which consists of a forwarding plane, signaling and control plane, and data plane.                                                                                                                                                                                                                                                                |
| Sankaran, 2023 [4]                                                                            | IPSec                                                 | A virtual network was set up to perform the experiment and a cross-protocol attack was performed on the IPSec tunnel. There were encrypted ISAKMP and ESP packets which were discovered by Wireshark but no information of interest to an attacker was being provided. However, it has also been pointed out that the use of RSA certificates helped to protect against such attacks on VPNs. |

Table 3.4: Comparison of existing literature

## Chapter 4

# Findings Discussion

### 4.1 Discussion of the Research

This chapter delves into the findings derived from the research endeavours so as to be able to discuss solutions for the problem attempted at being solved. The chapter will begin by analysing the various research questions laid out while trying to demonstrate how they were answered or solved. This would be an extensive analysis discussing what were the findings of this research. This would be followed by the overall results analysis which will provide an overview of the insights which this work provided while comparing existing works as well. Finally, it will end with a summary of the key contributions and findings achieved through this research.

### 4.2 Analysis of Research Questions

The research questions shall now be tackled and they are as follows:

- **What are the different types of encryption used in various tunneling protocols such as GRE, IPSec and L2TP employed by VPNs? The analysis of the performance is also done for each technique used.**

Throughout this study, the different encryption mechanisms used in various VPN tunneling protocols such as GRE, IPSec, L2TP and L2TP/IPSec are analysed. An overview of what algorithms are used in some protocols was given earlier and this section delves deeper in the technical aspects of these and how different ones work better for specific kind of applications. The first VPN tunneling protocol analysed is GRE which is a site-to-site access protocol and statistics point out that 63 percent of organisations use this type of protocol to connect their branch offices [15]. GRE encapsulates the datagram of a given network protocol and then allows it to be transmitted in another network layer protocol such as IP. This is done through a tunnel and the encapsulation performed is generally robust but it does not provide data encryption on its own [40]. In other words, this protocol can be said to transmit IP packets or

non-IP packets through the public IP network [15]. This is one of the reasons why technologies such as GRE over IPSec are implemented so as to ensure security in data transmission through a VPN tunnel. This leads to the analysis of how the IPSec protocol works.

Considered to be a highly secure technology, IPSec makes use of a mix of encryption and tunneling techniques as well as authentication algorithms. These contribute in ensuring the safety of the packet payload when it is transmitted over the network [1]. AH and ESP protocols are used to provide authentication and integrity services and also confidentiality in the network layer. To provide confidentiality, ESP is utilised and it does so through a block cipher mechanism which runs in CBC mode. The whole inner IP datagram is encrypted and is included in the outer IP datagram's payload [26]. For the purposes of authentication and integrity of data, AH is used which in turn generate Hash Message Authentication Code (HMAC) for all packets transmitted. ESP when ensuring confidentiality through encryption does not consider the IP address when computing the HMAC and so, IP spoofing can not be done as data authentication has already been performed as soon as the tunnel was created [16]. To establish secret keys for securing IPSec traffic and for authentication reasons, IKE is used. In this protocol, ISAKMP is crucial as it provides a framework for the key management protocol [46]. This protocol also has 2 versions namely v1 and v2 and 2 phases which are main mode and aggressive mode. The latter mode includes six messages for IKEv1 between the VPN server and client and four messages for IKEv2 [21]. The IPSec VPN has proven to have several benefits when it comes to the enterprise network. First, it is economic as no expensive rent has to be paid for leasing fixed lines. It is also flexible as IPSec VPN can connect branches at any location. The protocol is also considered to be safe because of the security mechanisms it provides. It also offers a redundant design ensuring the reliability of the link and equipment by VPN devices. Finally, there is effective management provided by the split channel feature of VPN switches which offers the resources for visiting the Internet [40].

The next tunneling protocol analysed was the L2TP. This is in fact formed from the Point-to-Point Tunneling (PPT) from Microsoft and Layer-2 Forwarding (L2F) from Cisco. Main security features including encryption are not provided by L2TP itself. However, it can be used together with IPSec so as to secure it [12]. This is done by first encapsulating the packet in L2TP through various layers of encapsulation which are namely, L2TP, UDP, IPSec, IP and data link where IPSec is used for encryption purposes for the L2TP tunnels. The PPP payload (IP datagram) will be enclosed in a L2TP frame and then in a UDP frame. Then this message is further enclosed with an IPSec header which includes the AH and ESP [19]. This explains how the L2TP/IPSec use the IPSec together with 3DES/AES to secure a tunnel. This protocol also has several advantages. First, it works both on IP and non-IP networks; it has also been proven to support more than one protocol. There is more authentication protocols present such as RADIUS and tokens. There can be multiple tunnels at the same time for each user and finally, when it comes to IPSec NAT traversal, NAT compatibility is supported [9].

- **Is certificate-based authentication or a Pre-Shared Key better for security in IPSec?**

As seen earlier, to establish secure connections in IPSec, IKE and ISAKMP are used. Both of them provide authentication and key exchange algorithms to set up a secure transmission channel. The difference between the two protocols is that IKE exchanges encryption keys dynamically while ISAKMP lays emphasis on the negotiation and establishment of security associations [4]. These are types of pre-shared key (PSK) which are used and they can be run in either aggressive mode or main mode. PSK is generally used for device verification as servers have a significant requirement for digital signatures to be verified. The length of PSK is up to 128 bits in aggressive mode and it is constructed through a combination of Unicode characters. Given the high performance average cracking speed present nowadays, only 2 to 50 hours would be required to successfully break PSK for comparatively small PSK lengths [24]. In the main mode, ISAKMP and the ESP protocol form the PSK based on L2TP/IPSec. To compute the encryption key, K of Diffie-Hellman (DH), cookie, random value and security policy value work together to complete the DH key exchange. The ISAKMP protocol is also vulnerable to attacks and as the PSK is not known, it is crucial for the encryption key, K to be retrieved. After doing so, the PSK in the main mode can be cracked using the same method as in the aggressive mode [24].

In [4], an experiment involving a cross-protocol attack was carried out to demonstrate how certificate-based authentication can protect IPSec networks from such attacks. It was done by creating a virtual network consisting of two VMs and an IPSec VPN tunnel was established between the server and desktop VMs. Wireshark was used to intercept the ISAKMP and ESP packets. The capture discovered encrypted ISAKMP and ESP packets which are of no purpose to an attacker. Address Resolution Protocol (ARP) poisoning and ARP spoofing were used to carry out a MiTM attack in this work but these efforts were unsuccessful thanks to the RSA certificates and default security features of the IPSec protocol. The captured ISAKMP packets were examined to discover the vulnerabilities which might be present in the techniques used to perform the key exchange and authentication so as to perform a cross-protocol attack. However, the VPN was protected against such threats due to the presence of strong authentication methods and RSA certificates[4].

- **What are the attacks which target VPNs and how can they be mitigated?**

Some attacks to which VPN services are vulnerable to include MiTM attacks, DoS or DDoS attacks, replay attacks, DNS hijacking inter alia. In a MiTM attacks, communication is intercepted between two end devices connected through a VPN [4]. Attackers then launch exploits such as protocols exploit, decryption of known algorithms, dictionary and brute force attacks [7]. Experiments in relation to flooding-based DoS attacks were carried out in [33]. The emphasis in this work was on stateless flooding attacks which are dangerous as they are performed after IP spoofing has been done and thus, it is difficult to trace back the attacker. The main defense mechanisms against



DoS attacks include source blocking where requests are filtered based on their source. Rate limiting can also be used as an attacker will normally send requests at a higher rate than legitimate requests. A pre-shared key can also be used to authenticate where the request is coming from [33]. There has also been research carried out to simulate a DoS attack focused on the OSPF protocol adjacent route spoofing. To defend against it, the use of the Suricata IDS is proposed [30]. A relay attack can be said to be a MiTM attack as packets which had been captured earlier are retransmitted on the network repeatedly. This is considered as a dangerous one as it can also be executed within secure architectures and thus access can be gained to sensitive information through it [11].

Based on experiments carried out in [7], some main suggestions for preventing such attacks on VPNs have been made. First, the VPN should be updated with the latest firmware and up to date security patches should be installed. Logs should be kept for analysis and accounting reasons and unused services should be blocked. It was also proposed to implement multi-factor authentication (MFA) in VPNs. These were some of the main solutions recommended in [7]. However, there are other ways to better protect VPNs such as using a VPN with strong encryption protocol and a VPN with robust authentication method using either RSA certificates or a PSK.

- **How passwords can be acquired and how traffic can be decrypted in L2TP/IPSec?**

The main aim of a VPN is that data passing through the tunnel should not be visible to others. However, the work in [24] proposed a method on how the weaknesses of the L2TP/IPSec protocol can be exploited and how PSKs and passwords can be obtained. It was also seen how encrypted traffic could be decrypted through a MiTM attack. To be able to achieve so, the security vulnerabilities of the L2TP/IPSec are first analysed. According to a study performed in [10], 26 out of 30 commercial VPNs use a PSK for authentication purposes and this may be viewed as a vulnerability. L2TP/IPSec is amongst the protocol used in some of these VPNs. This protocol, however, offers two layers of authentication where the initial one is the use of IKEv1 and the second one is the use of the Password Authentication Protocol (PAP) or Challenge-Handshake Authentication Protocol (CHAP). Nevertheless, the preservation of confidentiality and integrity of the connection depends wholly on the IPSec tunnel as L2TP does not offer transport protection. Thus, it is analysed how the traffic in this tunnel can be compromised if there is access to a PSK. It has been seen earlier how the PSK can be cracked both in the main and aggressive modes. There are also vulnerabilities which can be scrutinized from two different angles namely the shortcomings in protocol design and those in implementation as well. When it comes to the protocol design, there may be particular attacks possible. For instance, the login password may be retrieved through weaknesses in the key exchange protocol. This can lead to attacks such as dictionary cracking, brute-force attacks and rainbow table cracking through a MiTM attack [24]. One way to prevent these attacks would be to start using certificate authentication instead of known PSK for authentication purposes [10].

- **How security in VPNs can be achieved with the minimum loss in performance?**

A challenge faced by organisations using VPNs is how to ensure that data is being transmitted securely with minimum compromise of performance as this is important for efficiency. To ensure secure connections, protocols such as IPSec, L2TP/IPSec and IKEv2 are used. Additionally, for confidentiality, encryption techniques such as AES and 3DES are used. Hashing methods such as SHA, MD5, PSK are used for integrity purposes and RSA certificates can be used for authentication reasons [1]. It is observed that the IPSec protocol uses different protocols such as AH, ESP, hashing algorithms, PSKs or digital certificates for security reasons. L2TP/IPSec uses the IPSec security mechanisms together with AES/3DES. However, as resources are needed for data encapsulation, the speed under this protocol is usually low. Performance of VPNs can be assessed using multiple factors. The work, [21] compared the performance of two VPN protocols namely the Secure Socket Tunneling Protocol (SSTP) and the IKEv2 by measuring throughput, jitter and delay. The experiment was performed on a Windows environment and it was seen that IKEv2 remarkably performed better than SSTP when packets were sent from one VPN client to another.

The key sizes of various encryption protocols is compared in [37] together with hashing algorithms as well. It was seen that for a 64-bit implementation, SHA-512 performs better in terms of throughput. It was also suggested to use SHA-256 together with AES-256 in IPSec tunnels for 32-bit implementations in the paper. The performance of SSL VPN was tested in, [41] in relation to encryption algorithms such as DES, EDES3, IDEA, AES and SCB2 under the Kylin operating system. The performance was measured using throughput and it was found that the transport protocol affects it more significantly than the encryption mechanisms used. Thus, cryptography can be used in specific tunneling protocols to further secure a VPN and also guarantee a minimum loss in performance in this case. The work, [38] carried out experiments to demonstrate how data plane solutions can contribute to improve the latency and throughput of IPSec traffic.

It has been seen that there are several ways to ensure security while also ensuring performance in VPNs. However, a trade-off will have to be made and this is up to organisations to decide which protocol to make use of in respect to their needs.

- **How traditional networking VPNs and cloud-based VPNs work?**

Traditional networking makes use of the Multi-Protocol Label Switching (MPLS) mostly so as hop by hop data can be forwarded after the exchange of label binding information has been done [28]. In other words as defined in RFC 3031, MPLS is a layer 2 technology utilised to enhance packet forwarding decisions in a conventional IP context [35]. When using this approach, IP addresses are not used to route packet information but rather labels are used. However, this has a high cost associated with the creation of VPN when using this technique which is why enterprises are considering to move in the direction of a cloud-based approach for the deployment of VPN services [28].

Most organisations which use VPNs would wish to see the VPN deployment costs go down in terms of operational costs, implementation costs and external costs associated with VPNs. The work, [8], studied a cloud model known as VPNaaS (VPN as a service) in an attempt to achieve this. The flexibility of business operations can be improved when using cloud computing services through its scalability and this contributes to reduce costs. It is done by an abstraction layer which is created to conceal the intricacies of the underlying hardware and separating the hardware and software [8]. When using traditional networking, new services required by clients are frequently retarded due to longer build times. However, in a cloud-based approach, this is possible to be done almost instantly to a large amount of user groups securely. This is because the cloud provider already has the necessary equipment to do so. This also illustrates how the use of cloud services can prove to be more efficient and resilient [28]. On-premise infrastructure can be connected with a cloud network when using a cloud VPN but there are still risks present when it comes to the matter of security. For instance, there may be impersonation attacks, packet sniffing and even key disclosure [25]. Therefore, it is seen that both traditional networking and cloud-based services VPNs have security risks associated to them but it depends on the aims and priorities of an entity which one would be more appropriate for its specific needs.

### 4.3 Overall Results Analysis

This section lays out a comprehensive analysis of the key insights which were obtained from this project which focused on the study of the security of VPN tunneling protocols. The objective of this work was to address the research questions outlined in Chapter 1 where the highlights were related to the encryption mechanisms used in tunneling protocols such as GRE, IPSec and L2TP/IPSec; what can be done so that the authentication process can be made more secure in VPNs and also the attacks to which VPNs are vulnerable to. It was equally analysed how cloud-based VPNs work compared against how VPNs which use traditional networking work. Some key findings have been made after conducting a systematic review of literature and these will be discussed in this section.

It has been seen that IPSec is a VPN tunneling protocol which provides various security mechanisms such as ESP and AH. However, it was also observed that when 2 protocols are combined, for instance, L2TP/IPSec, more security can be provided to users especially if there is a need for specific purposes. For example, L2TP can transmit IP packets over a non-IP network and then IPSec can be used to make the VPN tunnel safer by offering its confidentiality and authentication services. Tunneling protocols often use PSKs for authentication reasons but it was, however, pointed out how using RSA certificates could provide the same services and is more secure. This is because PSKs was vulnerable to being cracked and so using a certificate approach for authentication can also help in preventing cross-protocol attacks. Various attacks still target VPNs such as MiTM, DoS and replay attacks and mitigation strategies were proposed such as updating the VPN with latest firmware, using strong encryption, using IPS/IDS amongst others. It was further analysed how the traffic in the L2TP/IPSec protocol can be decrypted and how password acquisition can be performed

which compromise the security of the tunnel. However, solutions such as using certificates for authentication was proposed for this issue as well. It was seen how cloud-based VPNs work differently from those which use traditional networking. This illustrated how a cloud-based approach can have lower costs for organisations for the deployment of VPNs. It was analysed how robust security measures can be implemented in VPNs but at the same time taking into account the performance of VPNs. It was seen that there are different behaviours of performance against security under various operating systems and protocols used.

#### 4.4 Suggestions for Secure VPN Deployments

From the above analysis, the encryption and authentication algorithms used in different VPN technologies both using traditional networking technology and a cloud-based approach, there are suggestions which can be considered in order to have secure VPN deployments. Among individuals, the rate of VPN adoption may be viewed to be low and this is mainly because VPN apps were vulnerable to several security attacks such as traffic leakage, insecure VPN tunneling protocols and MiTM attacks [32]. Thus, this analysis will guide users, especially corporate organisations to make a choice of a VPN technology according to their respective needs.

As seen earlier, it is of utmost importance to ensure confidentiality when sensitive corporate data is being transmitted in a VPN tunnel. The first aspect to consider for this is the encryption mechanism used by a VPN. The one which provides the highest level of security should be chosen and it can even be the use of two combined protocols such as L2TP/IPSec. However, it must also be taken into account which type of application is being used through the VPN as different algorithms may function differently on various environments. It is also important to guarantee data integrity and this is done by considering the authentication mechanism used by the VPN. For instance, it has been seen that it may be better in most situations to use certificate-based authentication to defend the VPN against attacks. Many organisations will be reluctant to compromise the performance of the VPN in use with security. It was pointed out in the analysis that high security can be achieved in VPNs with a minimum loss in performance since it was seen that the performance differs on different types of environments such as different operating systems. The behaviour of VPN was also analysed and the benefits of using cloud computing were outlined. Implementing a cloud-based approach for deploying a VPN can prove to have lower costs, be more resilient and also be more scalable. This approach also provides security mechanisms at lower costs as a third-party is providing the services. However, this may also be considered a threat as there can be attacks such as packet sniffing and key disclosure. Therefore, the final decision to choose a secure VPN deployment will depend on the requirements of an entity and it may decide to use it either using traditional networking or on the cloud depending on the security level of the data being transmitted in the VPN tunnel.

## Chapter 5

# Conclusion and Future Works

### 5.1 Project Analysis

This report explored how various VPN tunneling protocols work in order to ensure connectivity. The study is carried out in the context of remote work and aimed at analysing the security in VPN tunnels to ensure that sensitive data is secure while being transmitted over a public network, that is, the Internet. This project is carried out mainly to encourage organisations to adopt the VPN technology and implement remote work systems. Research is carried out in the direction to explain what security mechanisms are offered by VPNs and at the same time, the vulnerabilities present were also scrutinized.

A meta-analysis is carried out to give a background on the topic and this included mainly the VPN tunneling protocols such as GRE, IPSec and L2TP. Existing research and solutions in the field are also presented and analysed. These are then compared and gaps were observed. The main ones are that there were no extensive work done yet on vulnerabilities in VPN and the encryption mechanisms used. Thus, the first research question involved analysing the encryption algorithms used in various tunneling protocols. It is pointed out how performance and security differed with different encryption mechanisms. It is also observed how protocols can be combined in order to offer better security. Another important aspect of security refers to the authentication services provided by VPNs. It is explained why it may be safer to use certificate-based authentication instead of pre-shared keys as these can be cracked. The vulnerabilities in VPNs are also studied which made them susceptible to attacks such as DoS, MiTM and data spoofing.

A research question was based on how traffic can be intercepted and also be decrypted by an attacker. As a result, passwords can also be acquired. It was seen that this was possible if the pre-shared key was cracked especially due to weaknesses in the key exchange protocol. An important point addressed was the trade-off between security and performance in VPNs. However, it was seen that these behaviours differ under varying circumstances. As network services are also provided by third-party providers, the aspect of cloud-based VPNs should also be taken into consideration. The difference in the way that traditional networking VPNs

and cloud-based VPNs work were analysed. It was demonstrated how the latter could be less expensive and be more scalable.

It is, however, to be noted that this project initially started as an implementation-based project. It consisted of developing a cloud server through OpenStack and Kali Linux to simulate attacks and assess the security of the VPN tunnel created. Ubuntu was used and the OpenStack dashboard (Horizon) was successfully set up as shown below.

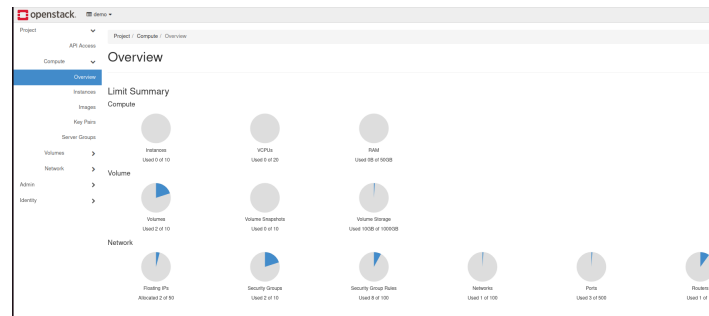


Figure 5.1: OpenStack Dashboard

However, technical issues were encountered while creating instances on the platform despite troubleshooting steps taken due to which this experiment could not be carried out. Consequently, it was found to be better to switch the project from an implementation-based one to a research-based one. Nevertheless, the research questions were designed to address the security concerns in VPN and analysis was carried out with the aim of providing better security in VPNs.

## 5.2 Limitations of Project

This research focused on tackling research questions based on the security and vulnerabilities of VPN on different platforms, that is, traditional networking VPN and cloud-based VPN. However, data has only been collected from existing academic papers and no new experiments were carried out to better address the issues. This was mainly due to technical issues faced when trying to do so. A successful attempt at developing the VPN server on the cloud could have been more helpful to prove the security of VPNs and how they could be protected against attacks through experiments. No external data could be collected as well because of time constraints. Thus, it can be said that the various aspects regarding VPN security concerns in VPNs have been analysed but the research could also be improved.

## 5.3 Future Works

The scope of this research covers the aspects of VPN such as encryption protocols and how attacks can be prevented by using a secure authentication method. Further research could

include analysing the same issues but by using external data as well. It could also include more research on cloud-based VPN and what are the security risks associated to it. Experiments could also be carried out to set up a cloud-based VPN and simulate attacks in the tunnel in order to assess the security of this technology. For instance, there could be IPS and IDS systems implemented to secure VPNs against the vulnerabilities analysed. There could also be the use of MFA to improve the authentication mechanisms so as to better protect VPNs against attacks such as MiTM attacks where passwords are decrypted and passwords are also acquired by the attacker. Firewall could also be used in order to filter traffic which may prevent DoS attacks. These could be done by using a VPN server on the cloud as it also means better management of resources. Further research could be performed in the direction of how security mechanisms in traditional networking VPNs could be integrated with security features of cloud-based VPNs as well and the concept of SDN could be studied in more depth.

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## .I Appendix A

Do not amend before use


  
**Middlesex University London**

**Research Ethics Screening Form for Students**

Middlesex University is concerned with protecting the rights, health, safety, dignity, and privacy of its research participants. It is also concerned with protecting the health, safety, rights, and academic freedom of its students and with safeguarding its own reputation for conducting high quality, ethical research.

*This Research Ethics Screening Form will enable students to self-assess and determine whether the research requires ethical review and approval via the Middlesex Online Research Ethics (MORE) form before commencing the study. Supervisors must approve this form after consultation with students.*

|                            |                                                                              |                              |
|----------------------------|------------------------------------------------------------------------------|------------------------------|
| Student Name:              | Chetan Bissessur                                                             | Email: cd1497@live.mdx.ac.uk |
| Research project title:    | Securing Communication through Strong Encryption Algorithms in VPN Tunneling |                              |
| Programme of study/module: | BSc (Hons) Cyber Security & Digital Forensics                                |                              |
| Supervisor Name:           | Mrinal Sharma                                                                | Email: m.sharma@mdx.ac.mu    |

Please answer whether your research/study involves any of the following given below:

|                                                                                                                                                                                                                                                                                                                           |                              |                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------------------|
| 1. <sup>H</sup> ANIMALS or animal parts.                                                                                                                                                                                                                                                                                  | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 2. <sup>M</sup> CELL LINES (established and commercially available cells - biological research).                                                                                                                                                                                                                          | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 3. <sup>H</sup> CELL CULTURE (Primary: from animal/human cells- biological research).                                                                                                                                                                                                                                     | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 4. <sup>H</sup> CLINICAL Audits or Assessments (e.g. in medical settings).                                                                                                                                                                                                                                                | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 5. <sup>X</sup> CONFLICT OF INTEREST or lack of IMPARTIALITY.<br>If unsure see "Code of Practice for Research" (Sec 3.5) at:<br><a href="https://unihub.mdx.ac.uk/study/spotlights/types/research-at-middlesex/research-ethics">https://unihub.mdx.ac.uk/study/spotlights/types/research-at-middlesex/research-ethics</a> | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 6. <sup>X</sup> DATA to be used that is not freely available (e.g. secondary data needing permission for access or use).                                                                                                                                                                                                  | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 7. <sup>X</sup> DAMAGE (e.g., to precious artefacts or to the environment) or present a significant risk to society).                                                                                                                                                                                                     | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 8. <sup>X</sup> EXTERNAL ORGANISATION – research carried out within an external organisation or your research is commissioned by a government (or government body).                                                                                                                                                       | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 9. <sup>M</sup> FIELDWORK (e.g biological research, ethnography studies).                                                                                                                                                                                                                                                 | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 10. <sup>H</sup> GENETICALLY MODIFIED ORGANISMS (GMOs) (biological research).                                                                                                                                                                                                                                             | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 11. <sup>H</sup> GENE THERAPY including DNA sequenced data (biological research).                                                                                                                                                                                                                                         | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 12. <sup>M</sup> HUMAN PARTICIPANTS – ANONYMOUS Questionnaires (participants not identified or identifiable).                                                                                                                                                                                                             | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 13. <sup>X</sup> HUMAN PARTICIPANTS – IDENTIFIABLE (participants are identified or can be identified): survey questionnaire/ INTERVIEWS / focus groups / experiments / observation studies.                                                                                                                               | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 14. <sup>H</sup> HUMAN TISSUE (e.g., human relevant material, e.g., blood, saliva, urine, breast milk, faecal material).                                                                                                                                                                                                  | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |

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Figure 2: Ethics form


  
 Middlesex University  
 London

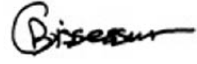
*Do not amend before use*

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                              |                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------------------|
| 15. <sup>H</sup> ILLEGAL/HARMFUL activities research (e.g., development of technology intended to be used in an illegal/harmful context or to breach security systems, searching the internet for information on highly sensitive topics such as child and extreme pornography, terrorism, use of the DARK WEB, research harmful to national security).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 16. <sup>H</sup> PERMISSION is required to access premises or research participants.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 17. <sup>H</sup> PERSONAL DATA PROCESSING (Any activity with data that can directly or indirectly identify a living person). For example data gathered from interviews, databases, digital devices such as mobile phones, social media or internet platforms or apps with or without individuals/owners' knowledge or consent, and/or could lead to individuals/owners being IDENTIFIED or SPECIAL CATEGORY DATA (GDPR <sup>1</sup> ) or CRIMINAL OFFENCE DATA.<br><small><sup>1</sup>Special category data (GDPR- Art.9): "personal data revealing racial or ethnic origin, political opinions, religious or philosophical beliefs, or trade union membership, and the processing of genetic data, biometric data for the purpose of uniquely identifying a natural person, data concerning health or data concerning a natural person's sex life or sexual orientation".</small> | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 18. <sup>H</sup> PUBLIC WORKS DOCTORATES: Evidence of permission is required for use of works/artifacts (that are protected by Intellectual Property (IP) Rights, e.g. copyright, design right) in a doctoral critical commentary when the IP in the work/artifacts jointly prepared/produced or is owned by another body                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 19. <sup>H</sup> RISK OF PHYSICAL OR PSYCHOLOGICAL HARM (e.g., TRAVEL to dangerous places in your own country or in a foreign country (see <a href="https://www.gov.uk/foreign-travel-advice">https://www.gov.uk/foreign-travel-advice</a> ), research with NGOs/humanitarian groups in conflict/dangerous zones, development of technology/agent/chemical that may be harmful to others, any other foreseeable dangerous risks).                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 20. <sup>H</sup> SECURITY CLEARANCE – required for research.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |
| 21. <sup>H</sup> SENSITIVE TOPICS (e.g., anything deeply personal and distressing, taboo, intrusive, stigmatising, sexual in nature, potentially dangerous, etc).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <input type="checkbox"/> Yes | <input checked="" type="checkbox"/> No |

M – Minimal Risk; X – More than Minimal Risk; H – High Risk

If you have answered 'Yes' to ANY of the items in the table, your application **REQUIRES** ethical review and approval using the MOREform **BEFORE commencing your research**. Please apply for ethical approval using the MOREform (<https://moreform.mdx.ac.uk/>). Consult your supervisor for guidance. Also see *Middlesex Online Research Ethics* (MyLearning area) and [www.tiny.cc/mdx-ethics](http://www.tiny.cc/mdx-ethics) (CS students).

If you have answered 'No' to ALL of the items in the table, your application is Low Risk and you may NOT require ethical review and approval using the MOREform before commencing your research. Your research supervisor will confirm this below.

Student Signature:  Date: 27/10/2023

**To be completed by the supervisor:**

| Based on the details provided in the self-assessment form, I confirm that:              | Insert Y or N |
|-----------------------------------------------------------------------------------------|---------------|
| The study is Low Risk and does not require ethical review & approval using the MOREform | Y             |
| The study requires ethical review and approval using the MOREform.                      | N             |

Supervisor Signature:  Date: 27/10/2023

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Figure 3: Ethics form

## .I Appendix B

### Project Meeting Log CST3590

Student Name: Chetan Bissessur

Student Number: M00838543

| Date       | Comments                                                                                                                                                                                                                                | Supervisor's Signature |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 04/10/2023 | <ul style="list-style-type: none"> <li>Explained basic structure of how project proposal should look like</li> <li>Gave guidelines to start writing the project proposal</li> </ul>                                                     | HS                     |
| 14/10/2023 | <ul style="list-style-type: none"> <li>Advised to adapt a more professional style of writing</li> <li>Went through project proposal draft and explained what was missing</li> </ul>                                                     | HS                     |
| 06/12/2023 | <ul style="list-style-type: none"> <li>Feedback on literature review</li> </ul>                                                                                                                                                         | HS                     |
| 12/12/2023 | <ul style="list-style-type: none"> <li>Feedback for comparative analysis in literature review</li> <li>Went through summary of findings</li> <li>Talked on proposed solution</li> <li>Outlined how chapter 3 would look like</li> </ul> | HS                     |
| 11/01/2024 | <ul style="list-style-type: none"> <li>Discussed how to launch OpenStack</li> <li>Discussed different sections in chapter 4 (Development) and chapter 5 (Testing)</li> <li>Discussed structure of final report</li> </ul>               | HS                     |
| 01/02/2024 | <ul style="list-style-type: none"> <li>Tried setting up OpenStack &amp; debug</li> </ul>                                                                                                                                                | HS                     |
| 08/02/2024 | <ul style="list-style-type: none"> <li>Tried to set up OpenStack through a preconfigured VM but did not work (later worked on another laptop)</li> </ul>                                                                                | HS                     |
| 16/02/2024 | <ul style="list-style-type: none"> <li>Steps for development on OpenStack</li> </ul>                                                                                                                                                    | HS                     |
| 26/02/2024 | <ul style="list-style-type: none"> <li>Tried to fix issues with launching instance on OpenStack</li> </ul>                                                                                                                              | HS                     |
| 29/02/2024 | <ul style="list-style-type: none"> <li>Advised &amp; discussed the switch from implementation project to research project because of technical issues.</li> </ul>                                                                       | HS                     |
| 17/03/2024 | <ul style="list-style-type: none"> <li>Finalised research questions</li> </ul>                                                                                                                                                          | HS                     |
| 10/04/2024 | <ul style="list-style-type: none"> <li>Feedback on analysis of research questions</li> <li>Changes in literature review</li> <li>Feedback on research methodology</li> <li>Feedback on key findings and results</li> </ul>              | HS                     |
| 13/04/2024 | <ul style="list-style-type: none"> <li>Final review of project</li> <li>Discussed what to add in conclusion</li> <li>Discussed change of title</li> </ul>                                                                               | HS                     |

Figure 4: Project Meeting Logs